Exercise 9.1

Consider full conformal prediction with an exchangeable dataset and a symmetric score function. Instead of comparing the test score against all n training scores, compare it against a random subset indexed by $I \subseteq \{1, \dots, n\}$, sampled independently of the data:

$$\mathcal{C}(X_{n+1}) = \left\{ y \in \mathcal{Y} : S_{n+1}^y \leqslant \text{Quantile}\left((S_i^y)_{i \in I}; (1-\alpha)\left(1 + \frac{1}{|I|}\right)\right)\right\}.$$

Prove that this modified prediction set has marginal coverage at least $1 - \alpha$.

Solution Let $Z_i = (X_i, Y_i)$, $D_{n+1} = (Z_1, \dots, Z_{n+1})$, and define the scores at the true test response by

$$S_i = s(Z_i; D_{n+1}), \qquad i = 1, \dots, n+1.$$

Symmetry of s and exchangeability of the data imply that $(S_1, \dots, S_{n+1})$ is exchangeable. Indeed, for every permutation π of the $n+1$ indices,

$$(S_{\pi(i)})_{i=1}^{n+1} = \big(s(Z_{\pi(i)}; (D_{n+1})_\pi)\big)_{i=1}^{n+1} \overset{d}{=} (S_i)_{i=1}^{n+1}.$$

The equality uses symmetry in the dataset argument, and the equality in distribution uses exchangeability of the data.

Now condition on $I = J$, for any fixed nonempty subset J having positive probability, and put $m = |J|$. Independence of I and the data means that conditioning on $I = J$ does not change the distribution of the scores. Thus the m selected scores together with S_{n+1} remain exchangeable. Let

$$k_m = \lceil (m+1)(1-\alpha) \rceil.$$

If $k_m \leqslant m$, the quantile in the question is the k_mth smallest score among $(S_i)_{i \in J}$. The exchangeable-rank argument gives

$$P\{Y_{n+1} \in \mathcal{C}(X_{n+1}) \mid I = J\} \geqslant \frac{k_m}{m+1} \geqslant 1 - \alpha.$$

For completeness, break any score ties uniformly at random solely for this rank argument. The rank of S_{n+1} among the $m+1$ scores is then uniform on $\{1, \dots, m+1\}$. Rank at most k_m implies that S_{n+1} is no larger than the selected-sample quantile, which proves the first inequality even when scores tie. If $k_m = m+1$, the quantile is $+\infty$ by the conformal convention, and the conditional coverage is 1.

Averaging the conditional guarantees over all possible subsets yields

$$\boxed{P\{Y_{n+1} \in \mathcal{C}(X_{n+1})\} = \sum_J P(I = J) P\{Y_{n+1} \in \mathcal{C}(X_{n+1}) \mid I = J\} \geqslant 1 - \alpha.}$$

If an empty subset is allowed, define its threshold to be $+\infty$ and return $\mathcal{Y}$ on that event; the same conclusion follows. Notice that I need not be uniform among subsets, nor have a fixed size. ////